Theory of Emotion

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Agenda

1. Emotion Psychology
2. Emotion Models
3. Limitations of Traditional Affective Computing
4. Specificity of the emotions
5. Emotion and Brain Asymmetry

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Emotion Psychology

1. What emotions themselves are?
2. How emotions are elicited or generated?

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Emotion: A Definition

Emotion is a complex set of interactions among subjective and objective factors, mediated by neural/hormonal systems, which can
(a) give rise to affective experiences such as feelings of arousal, pleasure/displeasure;
(b) generate cognitive processes such as emotionally relevant perceptual effects, appraisals, labeling processes;
(c) activate widespread physiological adjustments to the arousing conditions; and
(d) lead to behavior that is often, but not always, expressive, goal, directed, and adaptive.

- *(P. Kleinginna and A. Kleinginna, 1981)*

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Emotions generation

- Do we run from a bear because we are afraid or are we afraid because we run?
- (James, 1894) proposed that we are afraid because we run.
  - Emotions are often accompanied by bodily responses (racing heart, tight stomach, sweaty palms, tense muscles, and so on).
- Fear is not directly caused by sense perceptions but by certain thoughts to which these perceptions may give rise (Worcester, 1893)
- The physiological responses return to the brain in the form of bodily sensations, and the unique pattern of sensory feedback gives each emotion its unique quality.
  - A well-armed hunter might feel joy | An ordinary man might feel curiosity

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Bidirectional Projections

- Brain Impacts on the body via visceral efferent pathways
- Body impacts on the brain through afferent feedback
  - *For Example: Laughter Yoga*
    - *Laughter is possible without humor or cognitive thought*
- Voluntary contraction of facial muscles contributes to emotional experience. (Strack et. al, 1988)
- Participants who hold a pencil with their lips, forcing their face to prevent or inhibit a smile, rate cartoons as less amusing than participants who hold pencil in their teeth, mimicking a smile (Strack et. al, 1988 , Levenson et. al, 1990).

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Emotion Models

- A distinction can be made between perceived and felt (induced) emotions (Gabrielsson, 2002)
- Perceived emotion is the emotion recognized in the stimuli.
- Induced emotion is the emotion experienced by the listener.
- Representation of emotions:
  - Discrete or categorical model and
  - Dimensional models

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Categorical Model

- In emotion recognition using classification methods, discrete emotion models account for the largest parts of studies carried out.
- Closest to what we use in our everyday lives when we use a single word to describe an affective state.
  - Darwin's evolutionary view of emotions (Darwin, 1872).
- Six basic emotions happiness, anger, disgust, sadness, anxiety, and surprise from universally recognized facial expressions. (P. Ekman and W. Friesen, 1971)
  - Contempt as a seventh pan-cultural facial expression.

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Categorical Model

- (+) From a computational point of view, these models are easy to implement
- (-) It is difficult to model relations between the discrete states.
- (-) Inconsistency is a source of criticism of these models, since it is not agreed upon which emotions are basic and which are not.

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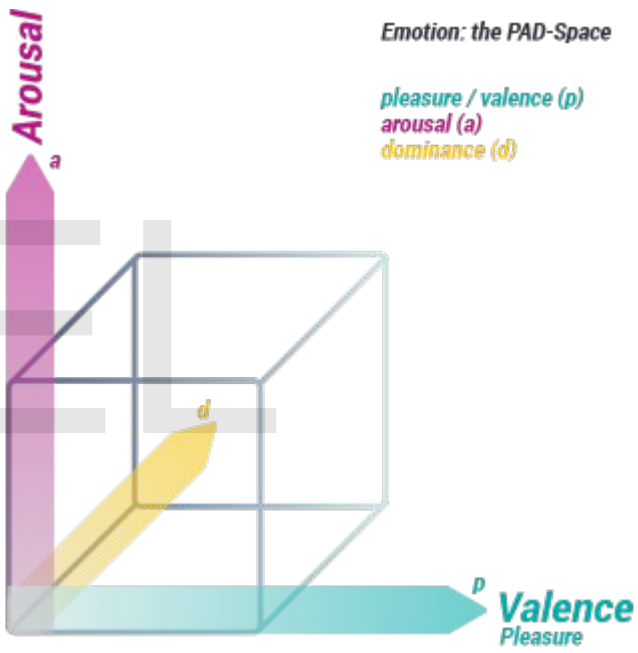
Dimensional models

- A 3D numerical vector denotes the location of an emotion within this space.
- VAD/PAD Model (Russell and Mehrabian, 1977)
  - Pleasure-Displeasure Scale: Pleasantness
  - Arousal-Non Arousal Scale: Intensity/Energy
  - Dominance-Submissiveness: Controlling and dominant nature of the emotion



Source: https://artplusmarketing.com/how-to-create-viral-content-with-studies-in-neuroscience-and-past-data-5e69c417adbe

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PAD model

- This model has gained attention in the field of emotion recognition lately, both in studies applying regression methods and studies that discretize dimensions into few areas.
- (+) Overcome the limitation of relating affective states to each other by providing a distance between them, i.e. they allow for computationally interpretable relations between emotional states.
- Often, the third dimension is omitted, resulting in the VA space: Circumplex model.

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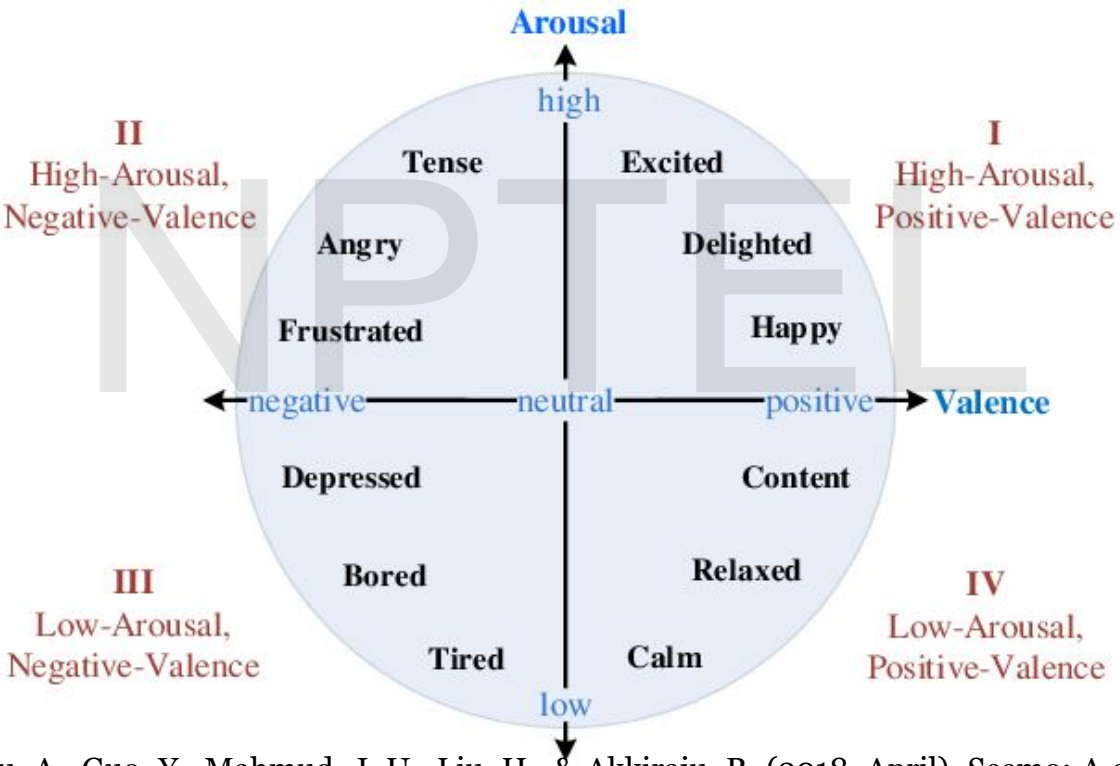
Circumplex Model



Source: Seemo: Liu, Z., Xu, A., Guo, Y., Mahmud, J. U., Liu, H., & Akkiraju, R. (2018, April). Seemo: A computational approach to see emotions. In Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (pp. 1-12).

Advantage of D in PAD model

- Fear can be characterized by
  - Valence: Negative
  - Arousal: Low/High
- Anger can be characterized by
  - Valence: Negative
  - Arousal: Low/High
- (-) We can not differentiate emotions that are overlapping!
- After including the D dimension:
- Fear
  - Valence: Negative,Arousal: Low/High , Dominance: Submissive
- Anger
  - Valence: Negative,Arousal: Low/High , Dominance: Control
- (+) Emotions that are overlapping in VA space, become distinguishable in VAD space.

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Problems in traditional AC

1. Human emotions include not only the emergency emotion stimulated by intense instantaneous stimuli, but also the process emotion stimulated by the accumulation of continued weak stimuli over a period of time.
2. Emergency emotions are quick and low-precision responses. The computational complexity of traditional precision-oriented affective computing is too high to handle emergency emotions.
3. Human emotions are personalized and the emotions of different human individuals excited by the same stimulus can be different.

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Specificity of the emotions

- Although basic emotions are characterized by specific facial expressions (Ekman & Friesen, 1975), a single set of facial actions can become different emotional expressions in different contexts (Barrett, 2012): Fear and Anger

Source: https://www.paulekman.com/universal-emotions/

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Fear

- Characterized by eyebrows raised and drawn together, wide open eyes, tense lowered eyelids and stretched lips. (Ekman & Friesen, 1975)

- Associated with activation within frontoparietal brain regions (Tettamanti et. al, 2012) and a broad pattern of sympathetic activation (Harrison et. al, 2013), including reduced heart rate variability (HRV).

- Also, associated with more numerous skin conductance responses and larger electromyographic corrugator activity than is anger (Stemmler et al., 2001)

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Anger

- Characterized by lowered eyebrows drawn together, tense lowered eyelids and pressed lips. (Ekman & Friesen, 1975)

- Left frontal PFC is activated (Harmon-Jones et al., 2010)
  - Literature is contradictory (Wacker et. al, 2008)

- No change in HRV (Rainville et. al., 2011)
- Anger may elicit either an anger-mirroring or a reciprocating fear response (Harrison et al. 2013)
  - Psychophysiological responses will be different.

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Disgust

- Characterized by raised upper lip, wrinkled nose bridge and raised cheeks. (Ekman & Friesen, 1975)

- Differential skin conductance responses may depend on whether the emotion is elicited by core-disgust inducing stimuli (e.g. Pictures of dirty toilets) or body-boundary violating stimuli (e.g. mutilation scenes)
  - Unchanged or decreased skin-conductance for core-disgust inducing stimuli, e.g. dirty toilets (Harrison et. al., 2013)
  - Increased skin-conductance for body-boundary violating stimuli, e.g. mutilation scenes (Bradley et. al, 2001)

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Sadness

- Characterized by raised inner eyebrows, lowered lip corners. (Ekman & Friesen, 1975)
- Associated with increased blood flow in ventral regions (Mayberg et. al., 1999).
- Crying related sadness:
  - Increased Heart rate
  - No change in HRV
  - Increased skin conductance
- Non Crying sadness:
  - Reduction in heart rate
  - Reduced HRV
  - Reduced skin conductance
  - Increased respiration

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Happiness

- Characterized by tensed lower eyelids, raised cheeks and raised lip corners. (Ekman & Friesen, 1975)
- The intensity of smiling in photographs has been found to predict longevity (Abel and Kruger, 2010)
  - Individuals with no smiles: ranged from 72.9 years
  - Partial smiles: 75.0 years
  - Duchenne smiles: 79.9 years
- Happiness is associated with activation in medial prefrontal and temporo parietal cortices (Tettamanti et al., 2012).
- A body of work further highlights the role of the left PFC in positive affects.
- Happiness is associated with decreased HRV, amusement and joy are associated with increases (Kreibig, 2010).

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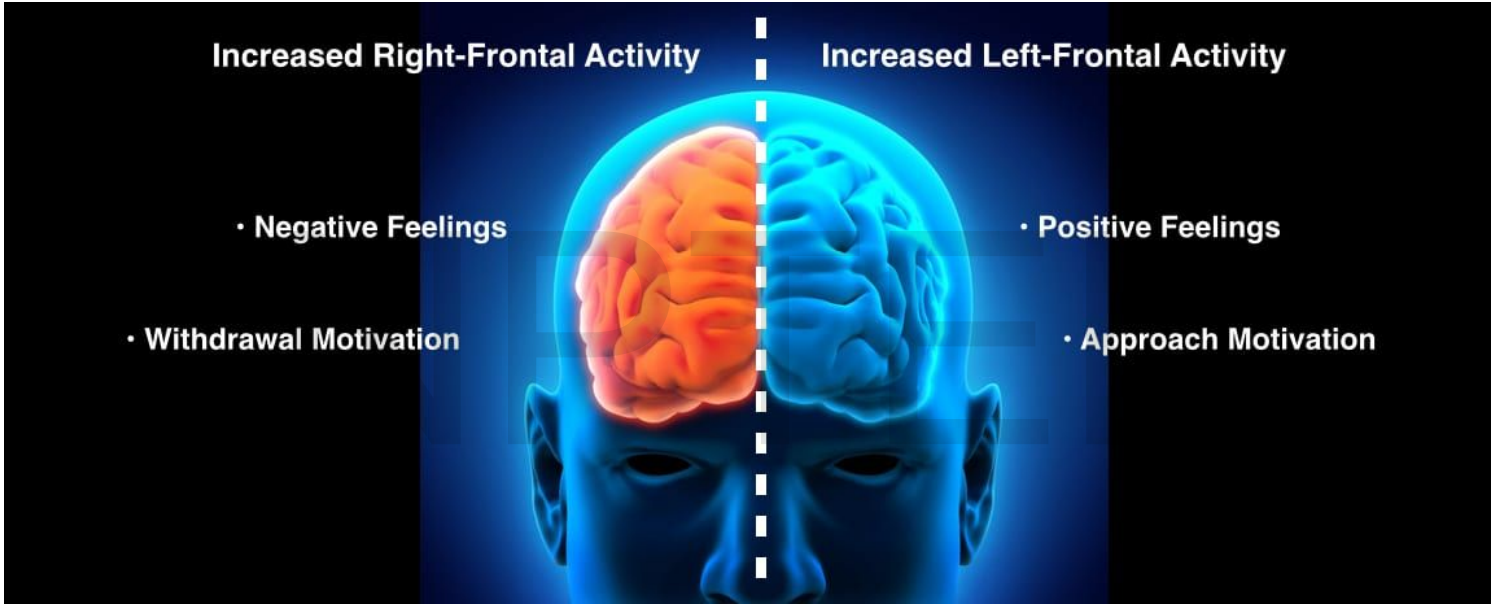
Emotion and Asymmetry



- Higher engagement of the left – relative to the right frontal brain – is related to positive feelings and higher engagement. (Coan et al., 2003).
- Exception Alert! Anger = Left Brain Bias (Harmon-Jones et. al., 1998)

Source: https://imotions.com/blog/learning/best-practice/frontal-asymmetry-101-get-insights-motivation-emotions-eeg/

Putting Emotions into Design

- How to create designs that evoke emotions which result in positive user experiences?
- Designers primarily focus on users' needs in their interactions.
- But similar focus on their responses is required, which are *naturally emotional*.
- Users have sophisticated thought processes going on most of the time.
- If you can elicit strong emotions in your users – you can use those emotions to either create loyalty or to drive a customer to take action.
  - *Emotional design* of a product or service affects its success.
- Designers aim to reach users on three cognitive levels—visceral, behavioral and reflective—so users develop only positive associations (sometimes including negative emotions) with products, services, etc.

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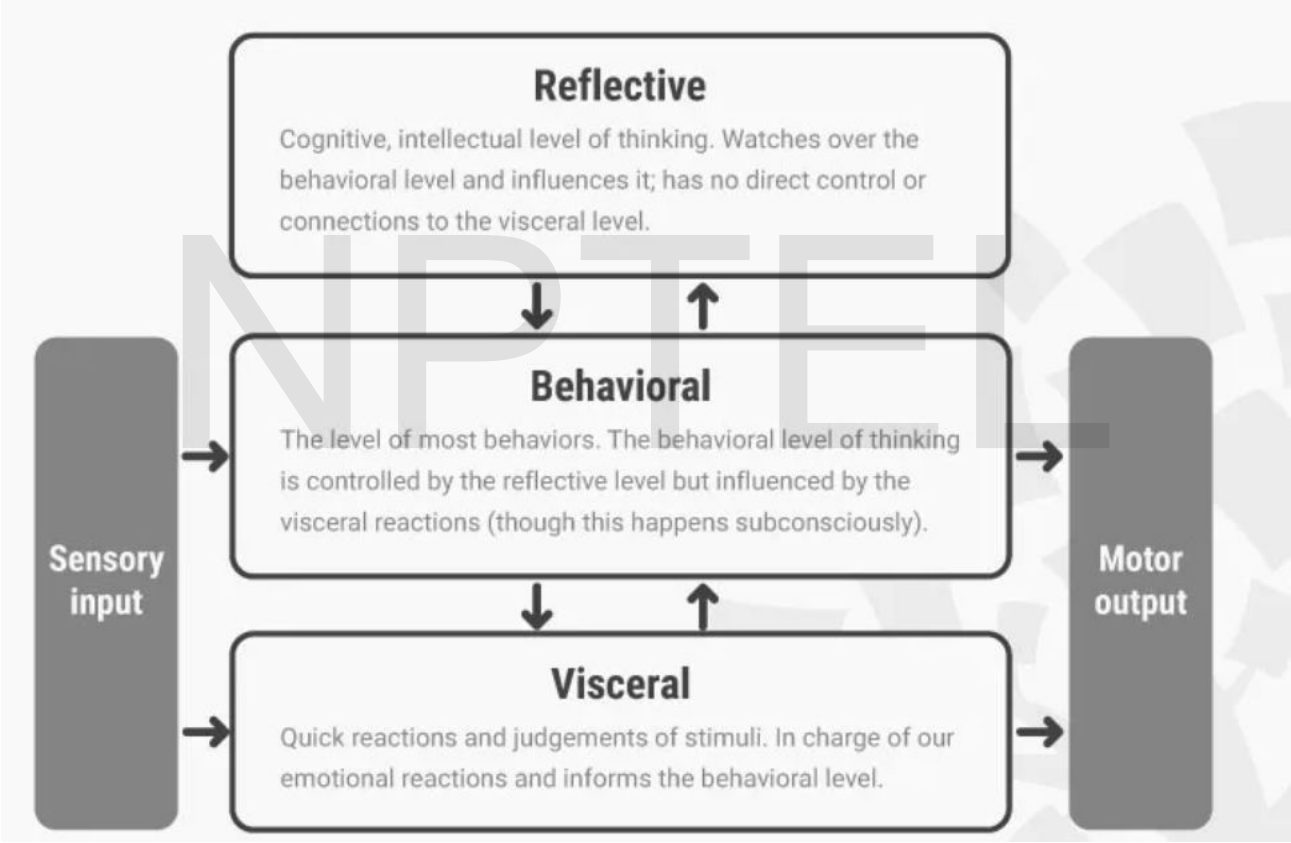
Three Levels of Design



Source: Emotional Design: Why We Love (or Hate) Everyday Things by Don Norman

Putting Emotions into Design

- The Visceral Design: Affective Computing can help you to understand the user's feelings in order to improve the first impression.
  - *First time you bought an iPhone*
- The Behavioral Design: Affective Computing can help you to understand the emotions the users would feel during the tasks; both successful and failed attempts. It can enable a thoughtful and emotional flow process within the design.
  - *Keyboard typing on MacBook*
- The Reflective Design: Affective Computing can enable better understanding of post-usage emotional experience and enable an emotional bond.
  - *Recall experience of a roller-coaster ride*

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